Engineering Physics: BAPHY105

Sequential Stern–Gerlach Experiment

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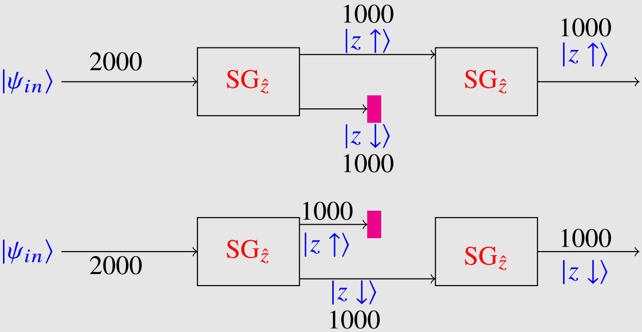
Lecture Outcomes

1. Sequential Stern–Gerlach Experiment

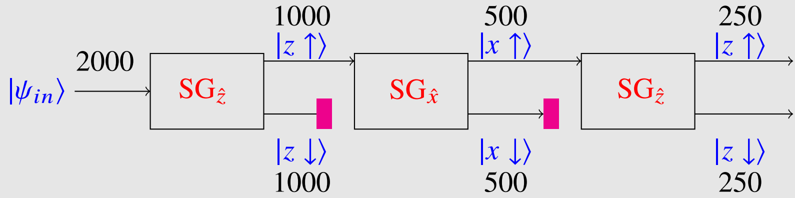
Sequential Stern–Gerlach Experiment

- ✓ The basic idea in sequential Stern–Gerlach experiment is to perform spin measurements along different axes sequentially.
- ✓ It helps us to analyse the impact of measurement of a physical quantity on the state of a system.
- ✓ The key question is whether measurement merely reveal a property or does it change the state??
- ✓ For a general mixed state ($|\psi_{in}\rangle$), the measurement of a physical quantity alters the state to one of the eigenstates of the associated Hermitian operator ($\hat{S}_z$). For spin$-1/2$ particles,

$$\hat{S}_z|\psi_{in}\rangle = +\frac{\hbar}{2}|z\uparrow\rangle \quad \text{or} \quad -\frac{\hbar}{2}|z\downarrow\rangle$$



- ✓ If the system is in a general state $|\psi_{in}\rangle$, then immediately after the measurement using the operator $\hat{S}_z$, the system collapses to one of its eigenstates: $|z\uparrow\rangle$ or $|z\downarrow\rangle$.
- ✓ If the system is in the eigenstate ($|z\uparrow\rangle$) of the corresponding Hermitian operator ($\hat{S}_z$), then the measurement does not alter the state. It continues to be in the same eigenstate after the measurement.
- ✓ Eigenstates of a physical quantity are mutually orthogonal.



- ✓ For spin−1/2 particles, S_x and S_z are two separate physical quantities whose eigenvalues cannot be specified together.
- ✓ There cannot be a simultaneous eigenstate for $\hat{S}_x$ and $\hat{S}_z$.
- ✓ $\hat{S}_x$ and $\hat{S}_z$ do not commute with each other.

$$[\hat{S}_x, \hat{S}_z] \neq 0$$

- ✓ The memory associated with $\hat{S}_z$ is erased by $\hat{S}_x$.